ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)

ACADEX PRACTICE EXAMINATION
O-LEVEL · MATHEMATICS

MATHEMATICS 4004/1

PAPER 1 November 2021

2 hours 30 minutes

PAPER 1 — SHORT-ANSWER — CALCULATORS MUST NOT BE USED

Additional materials: Electronic calculators must NOT be used. Geometrical instruments may be used.
INSTRUCTIONS: Answer all 30 questions. Show all working. The number of marks is given in brackets [ ].

1. Work out 2/5 + 2/3. Give a mixed number if possible. **[3]**

2. Evaluate 9 + 4 × 2 − 3. **[2]**

3. Overnight in Masvingo the temperature was 0°C. By 14:00 it was 20°C. Find the rise. **[2]**

4. A map of Masvingo uses scale 2 : 500000. A road is 10 cm on the map. Find the real length in km. (1 : 100 000 means 1 cm ↔ 1 km if 2:5 is first simplified — instead: scale 2 cm to 5 km.) **[3]**

5. In a class of St Mary's, n(A) = 11 (play soccer), n(A ∩ B) = 5 (play both). How many play soccer only? **[2]**

6. Find 25% of 80. **[3]**

7. Expand and simplify 6(x + 3) − 5x. **[3]**

8. Write 80000 in standard form. **[3]**

9. Simplify (3^4)^4. Leave as a power of 3. **[3]**

10. Find the HCF and LCM of 12 and 28. **[4]**

11. Write the equation of the line with gradient 3 passing through (0, 5). **[3]**

12. The nth term is $6 + 4(n - 1)$. Find the 8th term. **[3]**

13. A regular polygon has 5 sides. Find each interior angle. **[4]**

14. Factorise $x^2 + 7x + 6$. **[4]**

15. Make h the subject of $A = 2\pi r(r + h)$. **[4]**

16. In triangle ABC, $\hat{A} = 39°$ and $\hat{B} = 57°$. Find $\hat{C}$. **[3]**

17. Find the midpoint of (3, 7) and (3, 3). **[3]**

18. A triangular plot has base 6 m and perpendicular height 5 m. Find its area. **[3]**

19. Factorise $x^2 - 4$. **[4]**

20. u = (1 ; 1), v = (5 ; 5). Find 2u + v. **[4]**

21. A bag contains 5 red, 3 blue and 1 green beads. P(not green)? **[3]**

22. A ladder of length 5 m leans against a wall. The foot is 3 m from the wall. How far up the wall does it reach? **[4]**

23. Find the mean of 7, 6, 9, 10, 7. **[3]**

24. Area of a circle radius 21 cm. Take $\pi = 22/7$. **[4]**

25. A cylindrical drum has radius 7 cm and height 6 cm. Find its volume. $\pi = 22/7$. **[4]**

26. A shirt in Masvingo is marked $200. In a sale it is 25% off. Sale price? **[4]**

27. Solve $2(x + 4) = 18$. **[4]**

28. Solve $2x + 1 < 5$. **[3]**

29. In right-angled triangle PQR, $\hat{P} = 30°$ and hypotenuse PR = 16 cm. Find PQ, opposite 30°. **[4]**

30. Solve<<
>>x + y = 5<<
>>x − y = 3. **[4]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).